Mathematical introduction to Compressed Sensing

Lesson 1 : measurements and sparsity

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Organization of the course

Every Tuesday (1/02 – 8/02 – 15/02 – 1/03 – 8/03 – 15/03 – 22/03)

3 hours course (15:15 to 18:30) 15 minutes break.

No lesson (22/02)

Simulations using python + notebook + cvxopt + cvxpy (**install before**)

All the course material (python and course lecture notes) are available on my webpage

Evaluation:

- Python notebook or pdf report (see the details on my webpage)
- group of two students registration before 21/02 on my webpage in the comments section of my webpage
- Project defense between 28/03 and 8/04.

Aim of the course: analyze high-dimensional data

1. Understand low-dimensional structures in high-dimensional spaces
2. reveal this structure via appropriate random measurements
3. construct efficient algorithms to learn this structure

Tools:

1. (approximation theory)
2. probability theory
3. convex optimization algorithms

First lesson is about:

Three central ideas in this course:

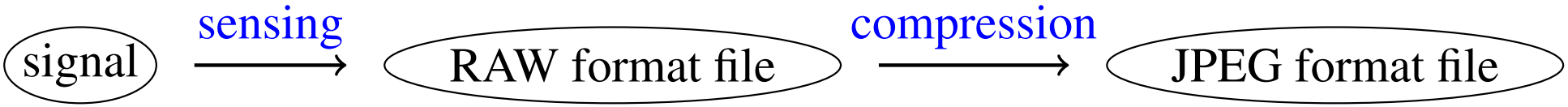
1. Structure = Sparsity, low rank, community, ...
2. random measurements
3. convex relaxation

In this lesson, we look at three problems which can be covered by a Compressed sensing approach:

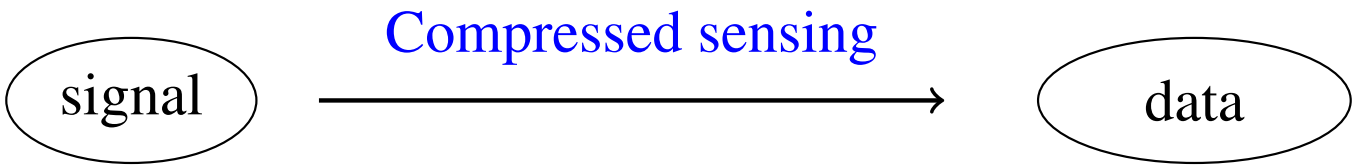
1. Single pixel camera
2. face recognition
3. Financial data

What is Compressed Sensing?

Classical data acquisition system in two steps:



Compressed sensing makes it in one step:



In french: Compressed Sensing = "acquisition comprimée"

Q: How is it possible? A: Construct clever measurements!

x: a signal (finite dimensional vector, say $x \in \mathbb{R}^N$)

Take m linear measurements of signal x:

$$y_i = \langle x, X_i \rangle, \quad i = 1, \ldots, m$$

where:

1. X_i : i-th measurement vector (in $\mathbb{R}^N$),
2. y_i : i-th measurement (= data = observation).

Problem: reconstruct x from the measurements $(y_i)_{i=1}^m$ and the measurement vectors $(X_i)_{i=1}^m$ with m as small as possible.

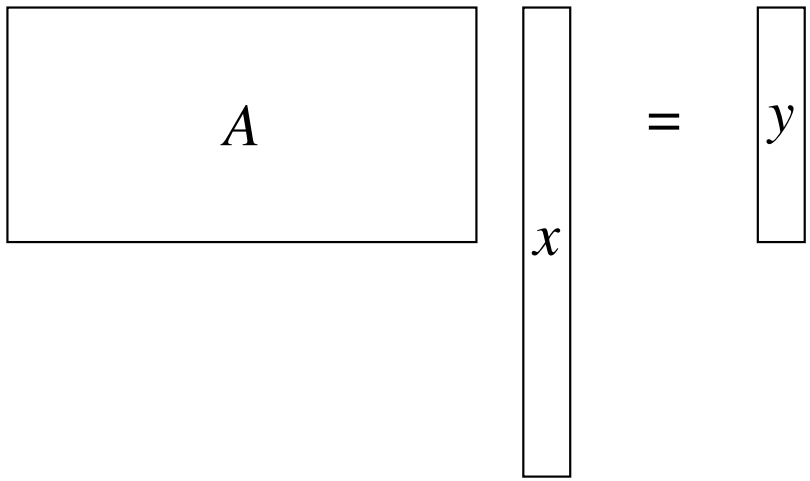
Matrix version of Compressed Sensing

We denote

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \in \mathbb{R}^m \quad \text{and} \quad A = \begin{pmatrix} X_1^\top \\ \vdots \\ X_m^\top \end{pmatrix} \in \mathbb{R}^{m \times N}$$

y: measurements vector and A: measurements matrix

Problem: find x such that $y = Ax$ when $m << N$



CS = solve a highly undetermined linear system

Sparsity = low-dimensional structure

Since $m < N$ there is no unique solution to the problem $y = Ax \Rightarrow$ no hope to reconstruct x from the m measurements $y_i = \langle x, X_i \rangle$.

Idea: Signals to recover have some low-dimensional structure. We assume that x is sparse.

Definition

Support of $x = (x_1, \ldots, x_N)^\top \in \mathbb{R}^N$:

$$\mathrm{supp}(x) = \{j \in \{1, \ldots, N\} : x_j \neq 0\}$$

Size of the support of x:

$$\|x\|_0 = |\mathrm{supp}(x)|$$

x is s-**sparse** when $\|x\|_0 \leq s$ and $\Sigma_s = \{x \in \mathbb{R}^N : \|x\|_0 \leq s\}$.

Sparsity and the undetermined system $y = Ax$

Idea: Maybe the kernel of A is in such a position that the sparsest solution to $y = Ax$ is x itself?

Procedure: Look for the sparsest solution of the system $y = Ax$:

$$\boxed{\hat{x}_0 \in \underset{At=y}{\operatorname{argmin}} \|t\|_0} \tag{1}$$

which looks for vector(s) t with the shortest support in the affine set of solutions

$$\{t \in \mathbb{R}^N : At = y\} = x + \ker(A).$$

Idea: Denote $\Sigma_s = \{t \in \mathbb{R}^N : \|t\|_0 \leq s\}$. If $\Sigma_s \cap \big(x + \ker(A)\big) = \{x\}$ for $s = \|x\|_0$ then the sparsest element in $x + \ker(A)$ is x and so $\hat{x}_0 = x$

Definition

$\hat{x}_0$ is called the ℓ_0-minimization procedure

(cf. Second lesson)

Compressed sensing: problems statement

Problem 1: *Construct a minimal number of measurement vectors* $X_1, \dots, X_m$ *such that one can reconstruct any s-sparse signal* x *from the* m *measurements* $(\langle x, X_i \rangle)_{i=1}^m$.

Problem 2: *Construct efficient algorithms that can reconstruct exactly any sparse signal* x *from the measurements* $(\langle x, X_i \rangle)_{i=1}^m$.

Is signal x really sparse?

Sparsity of signal x is the main assumption in Compressed Sensing (and more generally in high-dimensional statistics).

Q.: Is it true that "real signals" are sparse?

Three examples:

1. images
2. face recognition
3. financial data

Compressed Sensing in images

Sparse representation of images



An image is a:

1. vector $f \in \mathbb{R}^{n \times n}$
2. function $f : \{0, \ldots, n-1\}^2 \to \mathbb{R}$

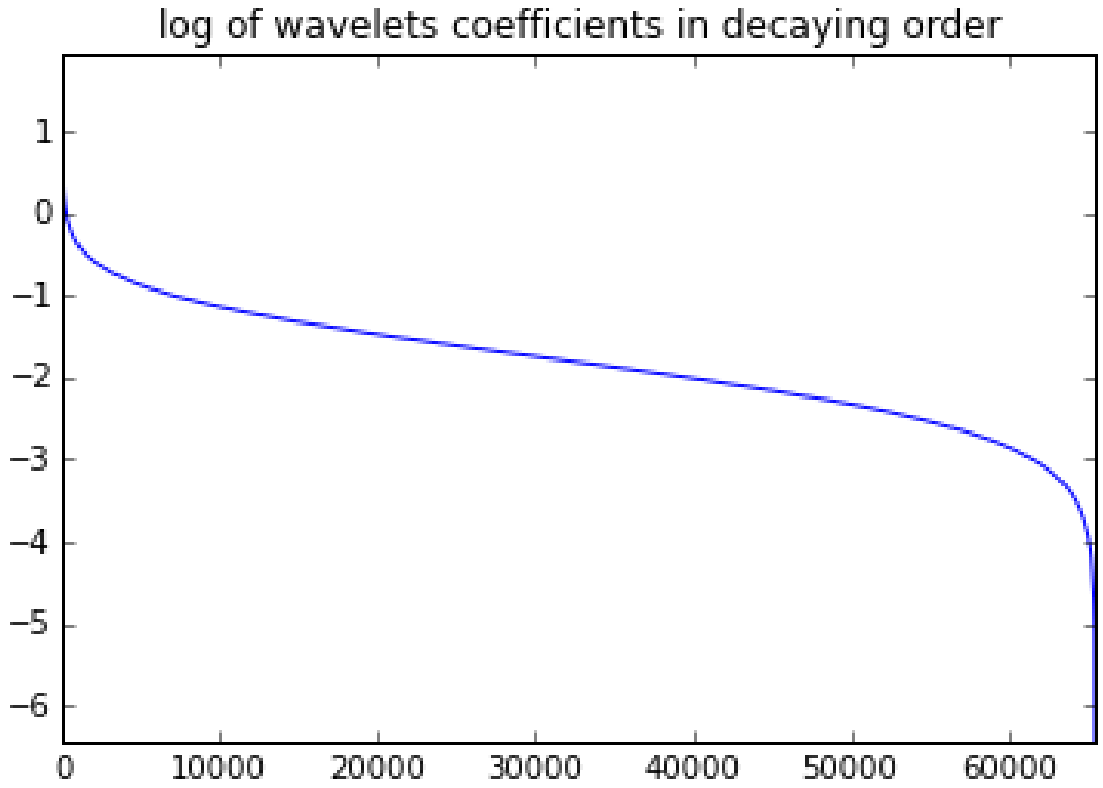
Images can be developed into basis: $f = \sum_{j=1}^{n^2} \langle f, \psi_j \rangle \psi_j$

Problem in approximation theory: Find basis (ψ_j) such that $(\langle f, \psi_j \rangle)_{j=1}^{n^2}$ is (approximatively) a sparse vector for real life images f.

Solution: Wavelets basis (cf. Gabriel Peyré course)

notebook: wavelet decomposition

Sparse representation of images



Graphics: Representation of $\left(\log|\langle f,\psi_j\rangle|\right)_{j=1}^{n^2}$ in a decreasing order for $n=256$ ($256^2=65.536$ coefficients).

Conclusion: When developed in an appropriate basis, images have an *almost* sparse representation.

Sparse representation of images





Idea: Compression of images by thresholding small wavelets coefficients (JPEG 2000).

Remark: these are the only three slides about approximation theory in this course!

Compressed sensing and images

Two differences with the CS framework introduced above:

1. images are almost sparse
2. images are (almost) sparse not in the canonical basis but in some other (wavelet) basis.

Two consequences:

1. our procedures will be asked to "adapt" to this almost sparse situation: stability property
2. we need to introduce a stuctured sparsity: being sparse in some general basis.

Structured sparsity

Definition

Let $\mathcal{F} = \{f_1, \cdots, f_p\}$ be a dictionary in $\mathbb{R}^N$. A vector $x \in \mathbb{R}^N$ is said **s-sparse in $\mathcal{F}$** when there exists $J \subset \{1, \ldots, p\}$ such that

$$|J| \leq s \text{ and } x = \sum_{j \in J} \theta_j f_j.$$

In that case,

$$x = F\theta \text{ where } F = [f_1 | \cdots, |f_p] \in \mathbb{R}^{N \times p}$$

and $\theta \in \mathbb{R}^p$ is a s-sparse in the canonical basis of $\mathbb{R}^p$.
For CS measurements, one has:

$$y = Ax = AF\theta$$

where $\theta \in \Sigma_s$ and so one just has to replace the measurement matrix A by AF.
Conclusion: All the course deals only with vectors that are sparse in the canonical basis.

What is a photos machine using CS?

It should take measurements like:

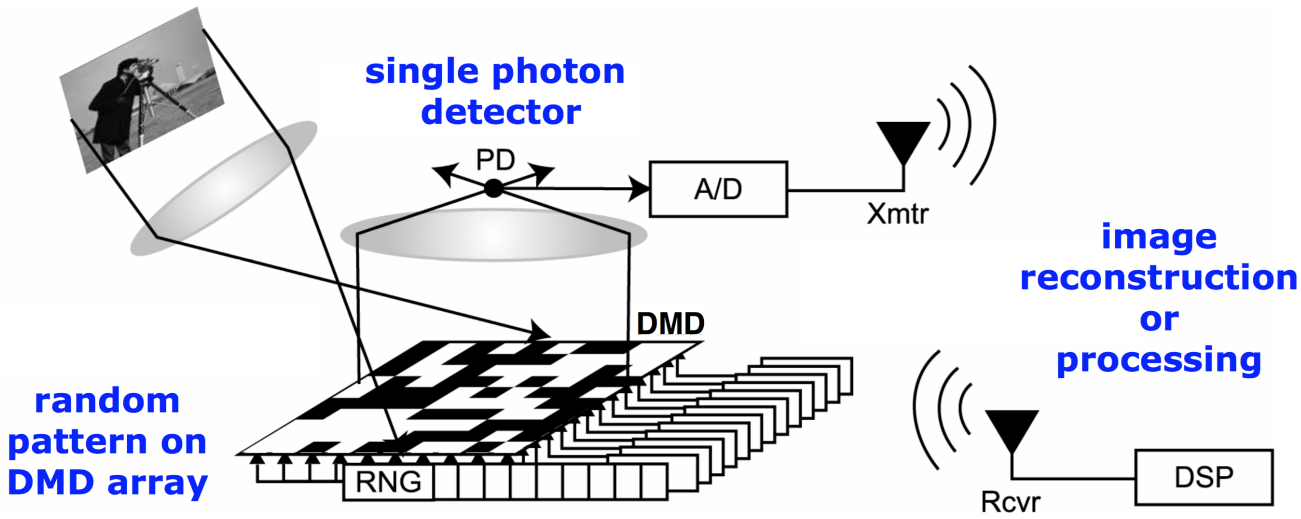


We take m measurements:

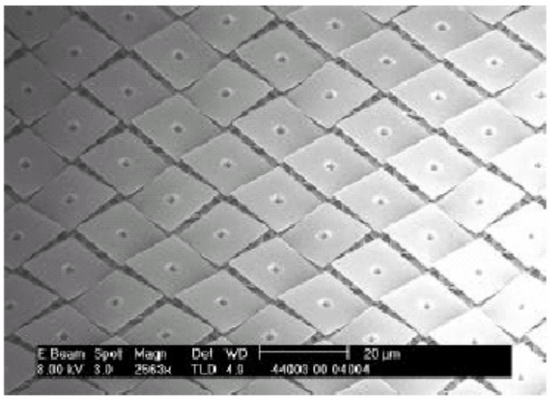
$y_1 =$  $, \ldots, y_m =$ 

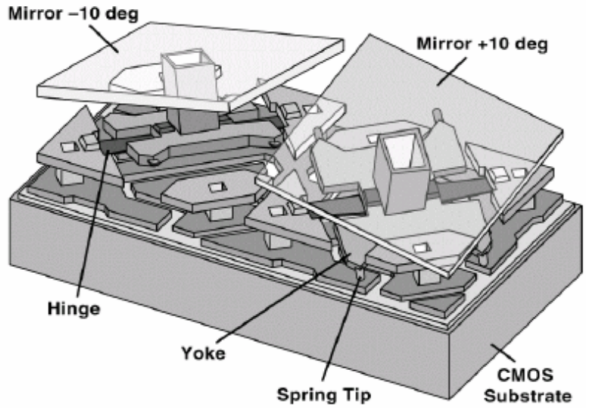
In particular, measurements $y_1, \ldots, y_m$ are real numbers. Each of them can be stored using only one pixel in the camera.

Single pixel camera from RICE University



DMD: digital micromirror device – randomly orientated





Single pixel camera from RICE University



Example of reconstruction of an image using the single pixel camera.

Two problems:

1. How do we choose the measurement vectors:  , $\cdots$,  ?
2. Is there an efficient algorithm to reconstruct the signal from those few measurements?

CS in face Recognition

face recognition and Compressed Sensing

Database: $\mathcal{D} := \{(\phi_j, \ell_j) : 1 \leq j \leq N\}$ where :

1. $\phi_j \in \mathbb{R}^m$ is a vector representation of the j-th image, (for instance, concatenation of the images pixels value)
2. $\ell_j \in \{1, \ldots, C\}$ is a label refering to a person

A same person may be represented in $\mathcal{D}$ several times from various angles, luminosity, etc..

Problem: Given a new image $y \in \mathbb{R}^m$, we want to label it with an element from the set $\{\ell_j, j = 1, \ldots, C\}$

"Classical" solution: use multi-class classification algorithm.

Here: Face recognition as a CS problem.

The sparsity assumption in face recognition

Empirical observation: If for all of the C individuals one has:

1. a large enough number of images,
2. enough diversity in terms of angles and brightness

then for any new image $y \in \mathbb{R}^m$ of individual number $i \in \{1, \dots, C\}$, one expect that

$$y \approx \sum_{j:\ell_j = i} \phi_j x_j.$$

Consequence: We assume that a new image $y \in \mathbb{R}^m$ can be written as

$$\boxed{y = \Phi x + \zeta}$$

where:

1. $\Phi = [\Phi_1 | \Phi_2 | \cdots | \Phi_C]$ and $\Phi_i = [\phi_j : \ell_j = i]$ for any $i \in \{1, \dots, C\}$,
2. $x = [\mathbf{0}^\top | \mathbf{0}^\top | \cdots | \mathbf{x}_i^\top | \mathbf{0}^\top | \cdots | \mathbf{0}^\top]^\top$ where $\mathbf{x}_i$ is the restriction of x to the columns of Φ_i in Φ
3. $\zeta \in \mathbb{R}^m$ error due to linear approximation of y by columns in Φ.

Face recognition as a noisy CS problem

Compare with the benchmark CS setup, one has three difference:

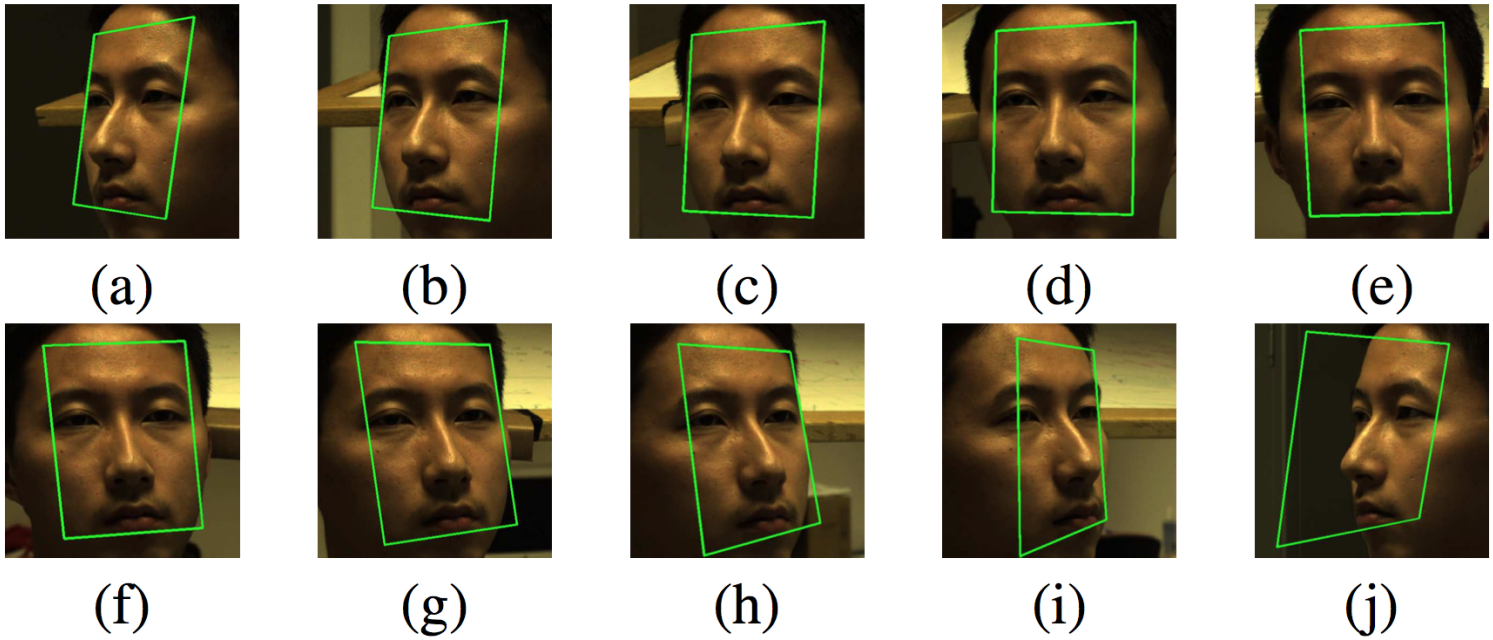
1. there is an additional noise term ζ
2. the sparsity assumption on x is stronger here: x is block-sparse
3. depending on the control one has on the database, we may or may not have the ability to choose (in a restricted way) the measurement matrix.

Three consequences:

1. our procedures will be asked to deal with noisy data: robustness property
2. we will design procedures taking advantages of more "advanced" sparsity like the block-sparsity one
3. when one is in a situation where there is no control on the choice of measurement vectors then one can try several algorithms and see how they behave.

Construction of a measurement matrix in face recognition

Various angles:



Various brightness:

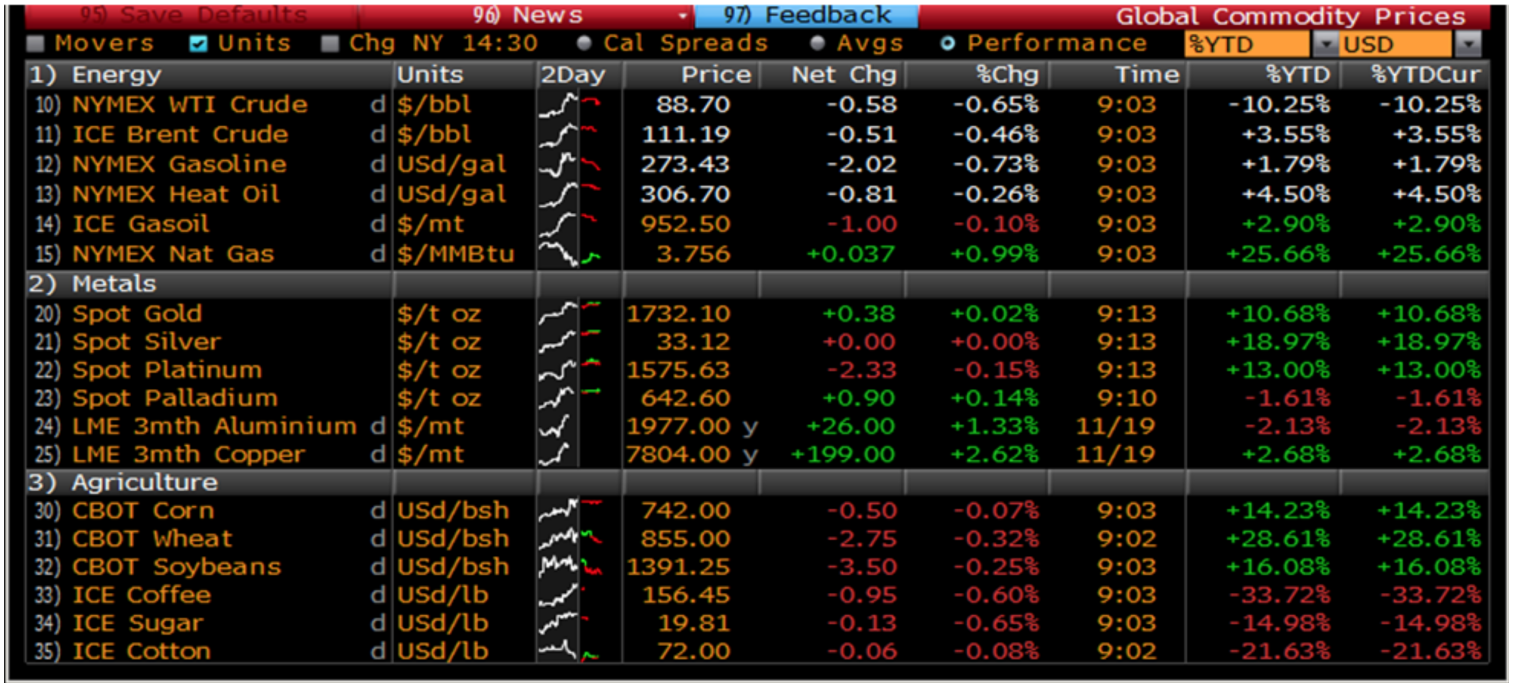


CS in Finance

Finance and CS

Problem: We observe the performances of a portfolio every minute: $y_1, \dots, y_m$. We would like to know how it is structured (shares and quantity).

Data: In addition to $y_1, \dots, y_m$, we know the values of all shares at any time:



95) Save Defaults | 96) News | 97) Feedback | Global Commodity Prices

Movers | Units | Chg NY 14:30 | Cal Spreads | Avgs | Performance | %YTD | USD

1) Energy		Units	2Day	Price	Net Chg	%Chg	Time	%YTD	%YTDCur
10) NYMEX WTI Crude	d	$/bbl		88.70	-0.58	-0.65%	9:03	-10.25%	-10.25%
11) ICE Brent Crude	d	$/bbl		111.19	-0.51	-0.46%	9:03	+3.55%	+3.55%
12) NYMEX Gasoline	d	USd/gal		273.43	-2.02	-0.73%	9:03	+1.79%	+1.79%
13) NYMEX Heat Oil	d	USd/gal		306.70	-0.81	-0.26%	9:03	+4.50%	+4.50%
14) ICE Gasoil	d	$/mt		952.50	-1.00	-0.10%	9:03	+2.90%	+2.90%
15) NYMEX Nat Gas	d	$/MMBtu		3.756	+0.037	+0.99%	9:03	+25.66%	+25.66%
2) Metals									
20) Spot Gold		$/t oz		1732.10	+0.38	+0.02%	9:13	+10.68%	+10.68%
21) Spot Silver		$/t oz		33.12	+0.00	+0.00%	9:13	+18.97%	+18.97%
22) Spot Platinum		$/t oz		1575.63	-2.33	-0.15%	9:13	+13.00%	+13.00%
23) Spot Palladium		$/t oz		642.60	+0.90	+0.14%	9:10	-1.61%	-1.61%
24) LME 3mth Aluminium	d	$/mt		1977.00 y	+26.00	+1.33%	11/19	-2.13%	-2.13%
25) LME 3mth Copper	d	$/mt		7804.00 y	+199.00	+2.62%	11/19	+2.68%	+2.68%
3) Agriculture									
30) CBOT Corn	d	USd/bsh		742.00	-0.50	-0.07%	9:03	+14.23%	+14.23%
31) CBOT Wheat	d	USd/bsh		855.00	-2.75	-0.32%	9:02	+28.61%	+28.61%
32) CBOT Soybeans	d	USd/bsh		1391.25	-3.50	-0.25%	9:03	+16.08%	+16.08%
33) ICE Coffee	d	USd/lb		156.45	-0.95	-0.60%	9:03	-33.72%	-33.72%
34) ICE Sugar	d	USd/lb		19.81	-0.13	-0.65%	9:03	-14.98%	-14.98%
35) ICE Cotton	d	USd/lb		72.00	-0.06	-0.08%	9:02	-21.63%	-21.63%

Finance and CS

$x_{i,j}$: value of share j at time i. We have the following data:

$$\begin{array}{lllll} t=1: & y_1: \text{ portfolio value} & (A_{1,j})_{j=1}^N: \text{ shares values} \\ t=2: & y_2: \text{ portfolio value} & (A_{2,j})_{j=1}^N: \text{ shares values} \\ \ldots\ldots & & \\ t=m: & y_m: \text{ portfolio value} & (A_{m,j})_{j=1}^N: \text{ shares values} \end{array}$$

Sparsity assumption: The portfolio contains only a limited number of shares and its structure did not change during the observation time.
Problem formulation: find $x \in \mathbb{R}^N$ such that $y = Ax$ where

$$y = (y_i)_{i=1}^m \text{ and } A = \left(A_{i,j} : 1 \leq i \leq m, 1 \leq j \leq N\right)$$

and x is supposed to be sparse.

CS and high-dimensional statistics

Definition

We say that a statistical problem is a **high-dimensional statistical problem** when one has to estimate a N-dimensional parameter / vector / object using m observations and $m < N$.

1. CS is therefore a high-dimensional statistical problem.
2. Noisy CS is exactly the linear regression statistical model when the noise and design matrix are assumed to be random and the parameter is assumed to be sparse.